

Jugaad Infrastructures: Platforming Habituated Ecologies during the Kerala Floods of 2018

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Abstract

Infrastructure has often been approached through the built, and yet hidden, substrates that facilitate modern life. In postcolonial contexts, the rise of platformized solutions tasked with performing infrastructural functions has introduced complexity in theorizing infrastructure. However, provisional and temporary solutions that bring together networks, relationships, and habitations as *jugaad* (informal or workaround) infrastructure have the potential to address crises beyond neoliberal capture enabled through platformization. This paper uses theories on alternative ways of assembling infrastructures to analyze a rescue portal built during the Kerala floods in 2018. It argues that these complex forms of assembling an infrastructure, even during the moments of imminent existential crisis, face the risk of platformized capture, surveillance, and governance.

Keywords

governance – infrastructure – India – *jugaad* – Kerala rescue – platformization – platforms

1 Introduction

Platforms have become a popular means for addressing infrastructural problems. In postcolonial contexts, they have thrived on the State's recent withdrawal from creating representational and symbolic structures on behalf of the nation. In India, this contravenes the post-independence push towards building national infrastructure intended to 'develop' the nation. Instead, platforms have become the go-to solution for concerns that merit large investment,

material and temporal. Quick fixes, which consider the digital superior, have emerged in the context of a global impetus to bring state-controlled infrastructure under privatized service delivery. The onus for governing populations is now on private entities, splinters of Silicon Valley – style conglomerates that have bid and bet on the latest infrastructure being constructed. In India, this new style of governance is already evident in the dashboard-based management of cities (Datta 2019), digital ID projects, portal-based delivery of welfare benefits (Khera 2017), and software solutions for the management of non(citizens) (Dagiral & Singh 2020). The move to subsume large societies digitally has been identified by some as the ‘financialized social’ that use a form of mandatory digital-mandatory financial service for day-to-day and essential transactions (Sundaram 2020b); others have focused on decisions aimed at bringing informal economic accounting under the ambit of service-based platformization (Athique & Parthasarathi 2020; Athique et al. 2018). Such conceptualizations are important when platforms have been outsourced not just for supplementing infrastructural needs but also for governing populations.

This article, however, presents a case study of an event in which platformization does not fit neatly in recent critiques. Keralarescue (KR), a portal built during the floods in Kerala (India) in 2018, has two oppositional sides. On the one hand, KR displays dynamic creativity, collaboration, and an effort to draw out sensory habituations – of body, affect, and technologies – and constitutes an infrastructure. In the words of AbdouMalik Simone (2004: 408), the portal can be read as a complex conjunction and interaction of ‘bodies, landscapes, objects and technologies’ that are yet to be formalized and institutionalized. However, on the other hand, KR risks capturing these networked innovations for corporate and capitalist profit making. Further, it adds to the catalogue of contextual tools that could facilitate platformized governance, as argued below.

2 A Unique Case: Why KR Stands Out

In August 2018, large parts of Kerala were flooded. As the state authorities were scrambling against time to manage the unexpected crisis, they experienced a concurrent infrastructural crisis. The dams of the state had reached capacity and, if water were not released, large portions of the state could have been inundated or broken under the stress (SANDRP, 2018). The physical infrastructural crisis was amplified by limited resources – at disaster management agencies, military and paramilitary – as well as a lack of coordination between different kinds of help, formal and informal. The challenge, as portrayed at the time, was

in strategically managing resources – human and material – which were being crowdsourced, largely through social media platforms. Keralarescue.in (KR) came about because of these concerns. Volunteers from engineering colleges across the state set up a portal to manage and organize life-saving operations in the immediate aftermath of the flood. Although other such efforts have been mobilized through social-networking platforms (including Facebook and Twitter), KR is a unique project in many ways. First, it had a unique ability to organize resources through a unified platform, collating rescue and relief requests that would have been lost in the din of social media posts and algorithmic feed curation. The volunteers arrived at a consensus that scraping rescue requests from across social media platforms was not possible because they comprised ‘noisy’ data: the requests were not organized into useful categories (name, phone number, address, etc.) or unifying hashtags, and, most important, were not organized based on location. This brings me to the second unique quality: its primary organizational feature was location gathering, which has its own set of logistical and political challenges, as elaborated below.

In some ways, the obsession with streamlining rescue and relief operations into neatly organized forms and automatic location pickers, was the start of the infrastructuralization of the portal. A concerted attempt was made to establish standards and protocols for data collection (Bowker & Star 1999; Edwards 2003). The portal quickly became popular and received the state government’s sanction, amplifying efforts to unify relief operations at a central node.

Third, and most important, KR stands out as an example of organizing a community with digital media tools in the face of a disaster. This includes the portal itself, which assembled an ecosystem of interactions and utterances mediated through a digital platform. It also involved the use of a host of other tools that supported the emergence of KR, including Slack, Github, and software APIs (application programming interfaces). Disaster notwithstanding, it is also part of a trend in which processes or tasks are undertaken by networks of community and kinship through dynamic innovation, both regular and provisional at the same time (Simone 2004), which are mediated by a platform. Rai (2020) calls the former mode of infrastructure *jugaad*, which is often called upon before, or in the absence of, infrastructure that emerges from formal resource organization.

This article situates and analyzes the work by KR that is part of this trend. I analyze how the formalization of temporary infrastructure falls under the logic of platformization as it is mediated through this portal. I begin by outlining the contours of platformization that are already evident in mediating essential infrastructure in India. In doing so, I trace the conceptual legacy of how terms ‘portals’, ‘platforms’, and ‘infrastructure’ are understood and how

these distinctions become murkier with the increase in digital socio-technical systems. I then explain the theoretical framework of *jugaad* infrastructure. For the moment, I define it as an ontological formation that assembles the productive multiplicities and potential of labour, with resources drawn from kinship relations, and sensorial and technological habituations, drawing on the work of Amit Rai (2020). However, I also juxtapose this with anthropological work on cultural techniques that emphasize physical and affective adaptation to technical media and how infrastructure in the Global South have a legacy of being provisional and temporary despite being dynamic, adaptive, and creative (Larkin 2014; Simone 2004). This leads the empirical and demonstrative part of my argument.

To do this, I draw on in-depth interviews as well as digital-ethnographic observations I conducted as an independent researcher in Kerala in 2018, a few months after the floods ended. These interviews were carried out with a few members of a nebulous team of over 2000 people responsible for building, designing, and maintaining the portal during and after the flood. The ethnographic observations were carried out on Slack, as I collated and examined relevant discussions on the Slack channel in which programmers, designers, community members, journalists, and representatives of national and global tech corporations, among other people, collaborated over long, often unstructured chats. They were open to the public after joining the group. The interviews were largely helpful for understanding how technical processes and decisions were made, as parsing technical terms over Slack conversations was beyond my expertise. Although I do not attempt to offer any insights or commentary on how specific technical decisions were made, I consider them in detail to form a holistic argument about the emergence of platformized infrastructure in this specific case.

3 Portal vs. Platform: Where Does KR Stand?

'Portals' have been known to provide customized experiences and interactions, characterized by strategic ways of organizing content, resulting in experiential uniqueness for each individual user. This is in line with how television scholars have theorized and understood them (Lotz 2017; Lotz et al. 2018).

KR is similarly embedded with functionalities (rescue form, database, donations, emergency information, etc.) that appear together in a single window: a portal for conveniently dealing with the disaster, facilitated by the web browser. The use of the term 'portal' is also particularly common in India as a solution or innovation in response to tedious encounters with bureaucracy (complaints

or grievance portals, a retiree portal, ticket booking portal, etc.). This implies a strategic rebranding of existing administrative or bureaucratic tasks mediated through a digital interface. In this case, KR as a 'portal' combines technical and design affordances (of data storage, location sensing, and multilingualism, among others), but it also aims to give individual users special attention as they are affected by a disaster, instead of becoming mere data points on a website or numbers affected by a tragedy.

In shifting from the use of the term 'portal' to 'platform' to describe KR below, I am mindful of the conceptual legacy of these terms, especially within numerous national and regional contexts and transfers. The semantic expansion of 'platform' is traced by Gillespie (2010) to strategic and business interests in the 2000s that attempted to morph the challenging work of being a cultural intermediary. To achieve this, the computational use of the term, combined with its architectural (a raised platform), political (place to speak and be heard), and figurative (enabling opportunities) uses to render it discursively resonant with neutrality, safety, and progressiveness, while being financially viable and visible enough for social appeal. In the East Asian context, Steinberg (2019) extends this analysis to argue that super-apps (e.g., LINE and WeChat) have moved away from the deployment of 'platform' to that of a 'portal' for another set of rhetorical and business motives, as they emerge with new functionalities embedded within chat apps.

KR exists somewhere in between these three iterations: (1) a subconscious absorption of the Gillespie-an platform model popular in the information technology (IT) industry; (2) resisting that model by bringing in different forms and types of platforming and eventually formalizing them, and, finally, (3) adapting the bureaucratic use of a 'portal' to be a disaster website (akin to super-apps in the limited sense of providing multi-functional services).

4 Platformizing Infrastructures: Trajectories and Emergences

This section lays down the theoretical underpinnings for how I approach infrastructure in this article. I start with the State's role in the development of physical infrastructure in India, and, as a result, its role in nationhood and citizenship.¹ I then discuss how more recent forms of mediated infrastructure have evolved in postcolonial contexts, locating them within global flows of

1 In what follows, I distinguish 'State' (with a capital 'S'), to discuss the state machinery and as a theoretical construct, from 'state' (with a lower-case 's') to refer to the Indian state Kerala, as an administrative unit.

people and technologies, indigenous ecologies of circulation and power, and the potential to escape overt State and capitalist control. This is followed by a review of how digital platforms have taken an infrastructural role, especially as they manifest changing meanings of 'platform' and 'infrastructure'. My goal is to develop a context in which KR, as a straddling of *jugaad* as well as a formal platform infrastructure, needs to be positioned and analyzed.

The Indian state, after independence from British rule, took the initiative in nation building, true to its socialist promise. This was primarily shown in the development of infrastructure (dams, highways, bridges, cities, industries, laboratories, hospitals, etc.). However, in the 1970s and 1980s, it was also extended to media and IT and became part of the national infrastructure built, coded, or designed. Media systems were designed for affective control, whether through sponsoring print or television infrastructure (Mankekar 1999; Sundaram 2015). The National Informatics Centre (NIC) was one of the key government organizations that was put in charge of developing the digital infrastructure on behalf of the government, among other public-sector companies such as Bharat Sanchar Nigam Limited (BSNL), which constructed the telecom infrastructure in the initial stages. However, in the 1990s because of large-scale disinvestment in government-run mega-establishments, along with a greater push towards private investment in essential services, the physical and mediated infrastructure was opened to private and corporate (re)configurations.

Digital, as the mode of organization for mediated and physical forms, came to India as part of these changes, which by this time had become a part of the neo-imperialist trade regulations. Sundaram (2015) argues that, because of low-cost technologies, shared devices and cheap personal mobile phones, networks that bypassed state and corporate power proliferated. These started to result in social and spatial transformations. However, these networks were also prone to attaching themselves to 'new platforms of political-aesthetic action' that found use for these floating and haphazard networks. In recent times, similar versions of networks have been expanded, reset, and reconfigured for platforming Hindutva nationalism through manufactured crises (Sundaram 2020a). This platforming is another metaphorical use of the space that can be used to consolidate relations and affect over a shared agenda.

Scholars of information studies argue that digital platforms have also placed purposeful restrictions on sociotechnical gateways, a key feature of infrastructure in ensuring broad access by assembling heterogeneous systems (Plantin et al. 2018). Instead, their embrace of interoperability is facilitated by APIs (application programming interfaces). This, in turn, has led to unprecedented scaling, deals behind closed doors, and collaborations uncharacteristic of the 'freedom' afforded by the open standards of Web 2.0. Operating at such a large

scale has resulted in the creation of tech megacorporations that control the bases for many services (Athique & Parthasarathi 2020). Although this somewhat resembles the traditional understanding of infrastructure as large-scale systems of control (Larkin 2013), the only addition now is the reciprocal expectation of copious user data that fuels profits and makes this infrastructure inescapable. The opt-out function is also eliminated because of their ubiquity, sometimes through formal and mandatory uses.

For instance, the use of Aadhaar and associated interfaces have made it impossible to escape the rigidity of code for accessing welfare services (Kitchin and Dodge 2011). The goals of ending corruption and eliminating the whims of middlemen in service delivery are far from achieved (Khera 2017). Instead, the burden of repairing one's body and the harm to the subaltern subjectivity has been magnified (Mudliar 2020). These platform-based infrastructures have therefore been characterized by a transfer of social interest to the interests of corporate capitalists, as a decoy to the State's indifference and incompetence, and by ushering in a new regime of control built on platforms' unprecedented, interoperable, and infrastructural functions.

In India, the splintering of infrastructure has been captured by local Silicon Valley – inspired units, built indigenously in the region or in China despite recurring geopolitical tensions (Aouragh & Chakravarty 2016). Scholars (Dattani 2019; Hicks 2020; Jonnalagadda 2019) have identified tech conglomerates such as iSpirt have banded together for strategic interests exploiting the vast infrastructural potential of data interoperability and API regimes for expansions in sectors such as fintech, banking, and health care. IndiaStack, a vast collection of personal data from various sources (finance, health, consumption), is being conceived as infrastructure on which services are overlaid, similar to Google and Facebook's APIs. The State's facilitation of this enterprise, as an impetus to IT-enabled trade and boosting economic growth and surplus, is ironic when one considers the ideas of data nationalism that have simultaneously accelerated within the same discourse. It is in this context that KR needs to be placed and analyzed. Is it consistent with the infrastructural splintering and capture made possible by platforms? To what extent does it deviate from the infrastructuralization of platforms that could become potential tools for governance?

5 *Jugaad* as Infrastructure: Affect, Habituations, and People

Mobile technologies, within their proliferating multiplicities, have enabled networks that bypass state and corporate boundary keeping (Sundaram 2015).

These networks are embedded in informal practices of information storage and dissemination and pirate media ecologies, infrastructures used by the subaltern, particularly by caste, class, gender, and religious minorities, forest dwellers (*adivasis*), and other anti-majoritarian actors. Simultaneously, these networks also lend themselves to platform majoritarian formations. Recently, several scholars have tried to articulate these formations. Mukherjee (2019) discusses dissemination networks that come together through cheap storage devices, particularly a resurgence of oral and vernacular practices, and work-around cultures in Mewat (rural Haryana). Udupa (2018, 2019) and Nizaruddin (2021) have carefully outlined networks which have come about for propagating anti-Muslim and violent Hindutva nationalism. These networks, workarounds, practices, and potentialities are a hidden infrastructure that is also assembled every time it is needed. These could be articulations in a crisis or for fulfilling mundane, everyday tasks. However, these informal flows and multiplicities are not just collections of ecological networks from the past (kinship, friendship, community relations over time) and the present. They are also a manifestation of bodily, sensory, and affective experiences and habituations that have taken place over time.

Bodily and affective habituations are culturally situated; they reorient our interactions with technologies, because of cultural techniques deployed in using and adapting to them. In his anthropological account of techniques of adapting to loudspeakers in Nigeria, Brian Larkin (2014: 991) argues that bodily training to sustain or release attention, when encountered with loudspeaker sounds (or noise), is a function of religious, political, and cultural norms and 'the conscious and unconscious practical techniques humans develop to live in the world'. Although Larkin's analysis of affective and discursive experience engendered through training is analyzed in a specific Nigerian context, it resonates with this project on interrogating sustained practices and workaround adaptations. The attempt is to move away from an understanding of infrastructure that centres the materiality of the built environment or structures alone but includes regular and sustained ways of living, making, and becoming. In turn, they become ways or techniques around which social life (transactions and livelihoods) are organized, which Simone (2004) calls 'people as infrastructure'.

If there is one word for what is described here, it would be *jugaad*: an ontology that accounts for connections, flows, sources, kinship relations, and an intellectual and creative dynamism often at the level of an enterprising individual. Colloquially, *jugaad* is used in north Indian contexts to refer to workarounds that often circumvent formal routes and regular temporalities. In his book *Jugaad Time*, Amit Rai (2019) discusses it as a practice that yields an assemblage (of connections and relations, flows and multiplicities,

and 'virtual' potential of labour) and holds the potential for offering momentary respite from the 'debilitating regimes of universal capital'. Rai (2019, 2020), however, also warns that, although *jugaad* offers ways of effectively intervening in exceptional circumstances with experimentation and possibilities, it also reifies and commodifies the affective and productive capacities of the 'preindividual'.

In analyzing the instances and empirical examples from KR below, I am interested in both kinds of *jugaad*. The first, in which creative multiplicities of the preindividual and the affective ecologies are channelled towards creating, repairing, and managing a platform that attempts to mitigate the disaster. Second, with Irani (2019), I also read the portal's dynamic creation as a trend towards the commodification and market capture of *jugaad*, by a new class of entrepreneurial citizens in India, which is ultimately oriented towards knowing and governing a population.

6 Demonstrating *Jugaad*: KR's Many Platformings

This section begins the evidentiary section of the paper, including examples, anecdotes, interview excerpts, and other secondary sources that attempt to articulate the complexity of decisions involved in making the KR portal during the floods of 2018. Starting with examples that discuss the ideology involved in programming a portal for a humanitarian cause, it moves on to the role of ecological networks of support and the role of the portal in mediating and formalizing them.

6.1 *KR's Making: Programming Ideology within Media Habituations*

Kerala, a state in the southwest corner of the country, is renowned for its exceptional welfare indices compared with those in other states in India. Its economic surplus has largely been tied to its diaspora in different parts of the world and internal migrants in other states. Among them are IT professionals who combined their efforts with those of local engineers, students, and community members when the flood wreaked havoc in the state. I mention IT professionals and computer engineers in particular, because of the emphasis in this article on the KR portal. However, when considering the efforts to mitigate the disaster as a whole, it is important to discuss the role of fishers, local athletes, and manual labourers skilled at rock climbing, swimming, and rowing.

Simultaneously, care work was performed by nurses, religious orders in different faiths, call-centre workers, local businesses, and others. Local, national, and global networks combined their efforts with more formal support from

state (governments at the state and national level and military agencies) and global agencies (neighbouring countries, Red Cross, agencies affiliated with the United Nations and non-governmental organizations).

Although networks of support abounded, so did the need for help. At the origin of KR were people who felt the need to bridge the gap between them by bringing in systematic solutions, relying on technical and algorithmic processes. The portal was launched to coordinate rescue and relief requests from people. Soon, it expanded to include ways to categorize and prioritize these requests, deleting duplicates, coordinating relief supplies, managing volunteers, and relaying information to and from the state rescue agencies, among several other tasks. Building a functional portal within a few days requires creative ingenuity and dynamic innovation by programmers, designers, and a host of IT volunteers from around the world. The channelling of this energy was supported by several ideologies. In this case I use ideology as a way to explain the invisible sources of authority that informed some of the decisions made while the portal was built.²

Interviews with the programmers as well as government officials coordinating with the project mentioned free and open-source software (FOSS) as one of the big motivators. The work that scores of volunteers were instantly called on to do was in some senses radical and implied a degree of social change, irrespective of its intended outcome. The code they were writing was not to benefit a corporation or meant to be hidden away in a secret corner of the internet. Instead, it was driven by collaboration and teamwork. The use of platforms such as Github, soliciting requests from virtually anybody, was a part of building the technical structure of KR, intentionally blurring lines of organization and hierarchy. Despite having to coordinate with the state government's IT efforts, the idea was to keep the 'spirit of volunteerism' undisturbed, a government official with whom I spoke told me.

Open-source software also dictated some of the initial technical decisions about the portal. In particular, Open Street Maps (OSM) was tried out as a mapping solution to mark flooded areas and inaccessible streets and for use as a base map on which other utilities were overlaid. This was thought crucial for

2 Ideology here is being used in the way that Stuart Hall deploys it based on his explanation of Antonio Gramsci's understanding of the term. Gramsci, according to Hall, addresses organic ideologies that are deployed to 'organize human masses and create the terrain on which men move, acquire consciousness of their position, struggle, etc.' (Gramsci, in Hall 1986: 43) This use of ideology here is different from the materialist philosophy debates about the term. It is, in fact, closer to Foucault's idea of discourse more than anything else, if one were to ignore the connotations of falseness from the former. For a detailed discussion of these ideas, see Hall (1986, 1992).

soliciting location-based rescue requests. However, it was immediately rolled back because of people's lack of familiarity with using the gestural functions in OSM. Google Maps, however, offered functionalities and design with which more users were familiar because of its incredible popularity. This was often the case during KR's development. Although the initial solutions came from the free and open-source software, migrations soon had to be made to more popular and corporate software because of a lack of familiarity or the inability to scale up. This is what one of the programmers told me about the OSM-Google maps conflict during our interview in 2018:

But we wanted a more precise location., so we wanted the street address and everything. So, if we just say Kayamkulam, it is not good enough ... because we can't exactly filter out people from Kayamkulam, so we needed something which was more concrete, which would now pin-point.... you need a G-apps API key. You need a premium key to do this; I think the [free] limit is around 100, [so when] more than 100 people start to use it, the whole thing crashes – so that was also a major issue.

The other alternative was OSM. There was a member of the OSM in our team – they were working with the other map, ... when we implemented it, there was an issue. We lost like two to three requests maybe. We put it [on the site], and we immediately saw the problem and we rolled back.... We didn't want such issues to happen. If we hadn't seen that error, more requests would have gone without location.... We had to be very very sure that it was correct. That's why we laid off some pull requests because we wouldn't trade off, a little bit better data than no data.

VIGNESH, author interview, 2018

For Vignesh, the switch from OSM to Google Maps was a non-decision. A few unrecorded rescue requests were enough to motivate a switch to a paid and expensive platform. Even though a premium key to the Google Maps API was eventually acquired through a volunteer, it was a source of distress for the first couple of days. How could a crowdsourced portal, with little to no economic support, afford the cost of using Google? OSM has been used in previous and subsequent disasters, including the earthquake in Nepal (2015), the Ebola outbreak (2014–16), floods in Bangladesh (2017), and, most recently, the COVID-19 pandemic (2020) (HOTOSM n.d.). However, it was considered insufficient for the functionalities that the KR portal attempted to incorporate.

Deciding to opt for the use of the Google Maps API could have been read as an instance of conceding to the ubiquity of platformized infrastructure. However, creative and ingenious programmers found connections on the

Slack channel that enabled access the corporate plan. The choice to use Google Maps might not have been an ideal solution according to KR's overall ideological framework, and it seems informed by the exceptional situation. It ensures that requests for rescue were not left out of the databases that were being prioritized by disaster management authorities and other networks of rescue volunteers. The shortcomings of OSM in this situation also informed how cartographic work within open-source movements have been transformed since the 2018 floods (Express News Service 2020).

The habituation process, discussed by Rai (2020), elaborates how media practices are indigenized over time from being 'strange' to being 'normal'. This could be a crucial way of thinking about why Google Maps API was adopted as the mapping solution for KR. As a technical tool whose use has been normalized over time, it has become a part of indigenous media practices, alongside other networks of *jugaad* infrastructure, which has led to this normalization (including people such as the mobile repair person or a community youth who assists in this familiarization). However, adding to Rai's reading of the indigenous media practices used by the subaltern, I maintain that the transgressive potential – of anti-capitalist and anti-Brahmanical resistance – can include people in coastal communities. As communities on the brink of existential threat and immense precarity because of climate change, coastal dwellers, especially in disaster-prone areas of the Global South, need to be recognized as the subaltern (Shome 2019). In that case, volunteer programmers and designers, the rescuers and boatsmen, the support network from around the country that is instantly mobilized – all become part of the on-call informal networks of *jugaad*. They are types of infrastructure that are assembled at a moment of crisis, drawing on links, relationships, and associations that have been built over time. The informal and provisional nature of these infrastructures extends to both bodies and their environment. The use of mobile technologies, apps, social-networking platforms, and their co-constituent materialities are all implicated within these habituations.

6.1.1 KR within Ecologies of Support

After the 2018 floods, images of fisherfolk carrying boats to carry out rescues from inundated homes became the disaster's dominant visual (Kumar 2018). The first responders were rock climbers, swimmers, and local fisherfolk who were solicited for help by personal networks, local leaders, and community elders. Artwork glorifying fisherfolk, carrying their boats on their shoulders to go to flooded areas, generated a tremendous outpouring of emotion for the unsung 'heroes'. However, responses in my interviews suggested that it became important to organize the local resources for a coordinated effort, instead of one that was hit or miss.

towards [the] 15th or something, we added a couple of categories like boatmen [and] rock climbers, because in Wayanad we had some specific needs [for] rock climbers and all. And in Pathanamthitta we had to add boat and boat services in Cochin. We also got some ... ex-navy people, some of [them] were also registered in ... that.

Anonymous government official, author interview, 2018

The portal became one of the ways in which these resources were organized and became part of the management processes: centralized organization and prioritized support. However, the interviewees to whom I spoke resisted classifying this as a way of formalizing that rendered resources exclusive. The community resources were not tabulated and categorized to prohibit them from helping people who were in need. Rather, the idea was to support and enable rescue operations through digital organizing tools. For instance, a key finding in the policy document prepared at the end of the KR project was that most houses in Kerala do not have addresses that could be directly coded as geolocations. A possible solution was to geocode them based on government-led projects, which could be an easier way of finding them during a disaster for rescue and relief purposes (KeralaRescue.in 2018). Because this was not possible during the floods, gathering copious location data while people were reaching out for help was deemed necessary to maximize the efforts at reaching and targeting them. In short, techniques deployed by portals such as KR have become ways to engage a community and assemble efforts at its own survival, along with collective mobilization resources during a crisis.

As mentioned earlier, *Jugaad* already involves a mode of production that commodifies and monetizes the work done by ecological infrastructure. It glorifies a neoliberal subject, the *jugaadu* (one who performs the *jugaad*), who can call on unequally distributed resources and mask it as creativity or dynamism. Despite this critique, it offers a negotiable paradigm for resource collection (the qualitative multiplicities) as opposed to the total monopolization that results from platformization. In that case, there is more merit in reading *jugaad* not a singular unified ontology but, instead, as an organizing technique that is situated in the indigenous practices of political, social, and cultural notions of power and, at the same time, offers ways to escape control from broad, overarching formalization and calcification that emerge from State or corporate infrastructural arrangements.

For instance, one way in which this dynamic process of power is manifested in the case of KR is through cyclical feedback that organically became a part of its design process. This feedback, rather than being part of an organized beta-testing process, in which participants are initially recruited, takes place through the networks in which the volunteer programmers are already embedded. The

resulting infrastructure, thus, is not manifested in monopolizing the process for profit but in generating creative appropriation, modification, and tinkering that emerge out of imminent and immediate responses by the people affected by the disaster. It takes place within the ‘historically specific affordances, capacities, tendencies, parameters of change, forces, machines, desires, values, and material flows’ (Rai 2020: 308).

Consider these two examples paraphrased from KR Slack posts:

At 10:47 PM on 16 August, a new user who joined the discussion group, said his ‘family’ had been stuck for the last 48 hours and were rescued by the Navy a ‘few hours ago’. From his networks he knows 1000s more are stuck in the same regions. He realizes that the region needs specialized attention because of the terrain it’s situated in, and needs to be prioritized. The solution he comes up with is to do data analytics, in order to prioritize these rescue requests. This is how the idea of building an algorithmic way of cleaning duplicate requests and assigning priority to certain requests comes about.

In a similar instance, on August 17, a ‘senior programmer’ shared a WhatsApp message at 4:24 AM about directing a 70-year-old person on how to register their request on the website. He goes on to ask the programmers to focus on the phone number as the only reliable information and that UX should not be confusing as the target audience is highly confused and bewildered.

Author summary of Slack discussions

These examples illustrate the affective habituations of the volunteer programmers, as members of communities determined to maximize their agency, who come up with modifications in the directions of programming solutions. Even though one can critique this for the privilege and power relations involved in who can afford to offer such suggestions, these corrections have implications for the portal’s design and organization, as well as subsequent actions. In the first example, algorithmic prioritization soon hit a roadblock because of the inability to accurately process data that was not clean or standardized during entry. This goes back to the earlier discussion on KR’s resistance to standardization but dynamic flexibility in programming operations.

However, another interesting use of habituated practice arises from this problem: call centres. Networks of volunteers, in other parts of the country and the world, are tasked with calling people’s phones (when possible) and to manually prioritize their requests based on the feedback they receive. This employs another set of habituations and resources and becomes crucial in reducing

duplication in the voluminous requests received on KR. In some cases, SMS alerts are sent to people requesting rescue or relief services so that they do not repeat the requests. Phone calls are used as a way to reassure people that their requests were heard and that help was on the way. These unorganized and dynamic call centres, which offer help verbally, derive from a model of customer service operations that has become normalized in the indigenous media ecology in India over the past two decades.

As an easy solution to mitigate the political ramifications of job losses in the West through ‘virtual migrations’ (Aneesh 2006) and in using affective labour practices (Mirchandani 2012), call centres became a major industry in India. A young and dynamic workforce willing to respond to excruciating, often racist, phone calls emerged as formal and pirate practices over the years. This was a different kind of platformization, one that abandoned collective local practices of knowing, sharing, and repairing for a monetized platform that offers directions and support facilitated by global telecommunications. However, over time the practice became part of people’s sensorial habituation that were crucial for generating new skills (the habituations that generate potential) based on sustained practices used in the call-centre model.

6.1.2 Dynamic Design: Choosing KR from among Platform Options

As the floods ravaged Kerala, volunteer programmers and designers were dismayed by the existing digital infrastructure developed in the heyday of State-sponsored digital infrastructure projects. As one of the research participants anonymously informed me that a portal already existed for the exact tasks that the programmers and researchers spent the first few days of the disaster building. However, the very fact that it was hosted on the National Informatics Center (NIC) servers made it impossible to maintain, access, or repair.³ Solutions were quickly sought on platforms used in other parts of the world for disaster mitigation. Two main platforms that were identified as likely candidates were Sahana Eden and Ushahidi. For all the talk of platform interoperability, they were developed based on the specificities of the regions where they were designed or were limited by language, address format, and even the software support they used. Ushahidi’s functionalities closely resembled those that the programmers were trying to obtain; however, the trick was in adapting

3 The National Institute of Disaster Management (NIDM) has designated several projects that are supposed to build such tools. India Disaster Resource Networks is one such consolidated portal that gives information about resources available at the district level. Many interviewees said that they did not hear about the use of any such portals during the entire duration of floods in Kerala in 2018.

it to the local context and its multilingual needs (residents of Kerala primarily speak and write in Malayalam). Similarly, modifying form categories, soliciting support from administrators in different time zones, and making changes at the drop of a hat in the design and functionalities of the portal were challenging. Ushahidi, once a close contender to KR, was scrapped as an option. Among other reasons, the Ushahidi team had withdrawn ‘free credits’ when they figured out that the portal was being used officially by the Kerala government for its rescue activities. Ushahidi, which evolved as disaster management software after its initial use in tracking post-election violence in Kenya starting in 2008 (Ushahidi n.d.), has been deployed in several natural disasters (including the Haiti earthquake) to gather user data and map locations.

It is not my purpose in this article to critique the funding model of non-profit organizations that work on humanitarian issues. Instead, I am interested in how the lack of dynamism afforded by these existing solutions made them ineffective when applied to Kerala. A decision also had to be made by the programmers after intense deliberation between volunteer programmers and corporate representatives (on the Slack group) about forgoing an existing platform (Google Person Finder) because of its limited functionality.

The volunteers’ decision to continue building KR from scratch led to functionalities that incorporated the local language and to creating fields for additional information (categories of data analysis that might not have been necessarily predicted). For instance, the previous example of requesting volunteers who were experts at swimming, rock climbing, or rowing shows the local specifics that needed to be included in the design considerations. If more rigid and standard software solutions been sought as the backbone of KR, these considerations would not have received the proper space and emphasis. This emphasizes my argument that KR is largely an evolving project that brought together the community’s needs with the resources that were made available over time. It was attentive to people’s physical needs and habits, training in technology and tools, language needs, topographic specifics, physical skills, demography, and other considerations. This is emblematic of the iteration of *jugaad* infrastructure of which KR is a part.

KR’s evolutionary design as a platform is in the form of a ‘list’ and avails of the dynamic possibility that lists as a genre accord when used in specific and contextual deployments, as opposed to aligning with universal standardizations. Lists, as opposed to a classification, recognize incompleteness (Adam 2008). ‘If someone comes along with something to add to the list, something that emerges as important, this may indeed be added to it’, Alison Adam remarks in relation to LISP (the now archaic programming language List Processor), which had this dynamic functionality as a part of programmability (Adam

2008). This is different from the earlier discussion of platforms and their programmability enabled by an API. APIs have locked-in standards, universal and unchangeable, that are replicated across programmed interfaces, including portals. Despite deploying an API for mapping and gathering locations, KR also had a space for typing in short distress messages built into its initial form filled in by those requesting rescue or relief services. This functionality was important, for instance, at a later stage, when attempts were made to prioritize some requests over others (e.g., for children, pregnant women, or the elderly). The volunteer programmers, rooted in the local context, were committed to adapting software solutions that were normalized over time as a part of habituated media (Google Maps, instead of OSM) but did not adopt and impose standardized solutions to an emerging scenario. The emphasis on formal innovation, prediction, and creative adaptation made space for *jugaad*, instead of creating a rigid code-space that cannot be circumvented.

As a portal that predicates its design and functioning on the principles of dynamism, solicits feedback from its potential users, and keeps its structure amenable to change, KR became a platform in an atypical sense. It appropriated habituated memories and everyday practices from other platforms and incorporated them to address a momentary crisis. It is involved in the API regime of outsourcing services and data sharing, yet attempts to avoid being subsumed by a tech corporation or being platformized by a non-profit organization. KR, then, represents a platforming of the people in Kerala and their affective networks and associations. It assembles skills and habituations, whether material, bodily or digital, with an ecology of support: parallel technical operations, such as fundraisers, Facebook groups, Twitter feeds, call centres, donations, prayers, debates, environmental consciousness, and administrative action. This is the plurality of platforming in which KR is involved. KR stands out from the strategic or managerial understanding of a platform by consciously not positioning itself as the ultimate solution based solely on technical expertise.

7 Commodification and Governance: The Flip Side of Formalizing *Jugaad*

This section offers a different perspective on the *jugaad* narrative. Instead of simply observing KR as a dynamic platform solution that vanishes when the disaster ends, it looks at the future of technologies or platforms such as KR and the model that it espouses. It also engages with popular imaginaries of general solutionism, which then open up community-led projects to co-option and commodification. A discussion of data storage, tracking, and surveillance

that follows is also particularly relevant to India as they are manifested in the government of marginalized populations on a regular basis.⁴ Although surveillance and governance are obvious frames for analyzing digitally mediated systems, I frame this section as speculative and cautious, in which the future of KR, or projects like it, is a topic in debates over the digital systems and platforms that have recently proliferated in India. I discussed this briefly at the beginning of this article.

Jugaad offers ways of assembling resources that resist a vertical exercise of power, expanding new and sustained practices of network formations, and incorporate cultural techniques for using and adapting to technologies. However, the impulse to formalize this risks commodification and nullifies the potential for resistance. Further, it involves *jugaad* in unintentional platforming (e.g., those for Hindutva nationalism) or towards the exercise of action and governance by the State.

Let us now consider the case of KR. As a socio-technical system, it has multiple degrees of care work built into it. This is not visible in digital systems and services that have patronage from the State. They include Aadhaar, the welfare delivery system (and, more recently, an identity verification system); the infrastructure built for the National Register of Citizens (NRC) in Assam for arbitrarily classifying (non)citizens; Aarogya Setu, COVID identification software that doubles as State surveillance; and CoWin, an infamously glitchy app built for vaccine registration. This is also the larger context in which software systems have produced insecurity, trauma, and precarity as a part of their operations. The infrastructure built through platformization in India is not just prone to breakdown but, rather, breakdown is a defining characteristic of it. So, apart from the incredible care work, dynamic repair, and community-based feedback cycles, KR has the potential to become part of a legacy similar to that of these platform-based infrastructures. That is when KR becomes a platform, as the term is understood as a strategic, monetized, and yet elusive intermediary, instead of simply a portal with temporary applicability. Solutionist models that tend to be replicated across contexts are the opposite of how KR as a project started out. However, my interviews and the future plans for the portal as articulated by senior programmers show that the process of standardizing KR and positioning it as a software-based solution for disasters in India and the rest of the world has already begun.

In her discussion of the entrepreneurial citizen, Irani (2019) predicts this. KR has the potential to be used as a tool of governance that is managed and

4 Smartphone apps were used to track the movement and control their phones remotely as a recent report on Asha workers (low-wage contractual health-care workers) evidenced (Bansal 2021).

implemented by the middle class on behalf of the State. According to Irani, 'empathy' and 'human-centered design' are often deployed as ways for entrepreneurial citizens to be in engagement with the people for whom they design and innovate. Irani's investigation, in a corporate design studio in Delhi, illustrates how 'design thinking' and 'entrepreneurship' are buzzwords used to produce a productive middle-class subject that works on behalf of capitalist interests. However, the State withdraws from social development (often infrastructure) and bestows its benevolence on activities that generate 'social surplus'. The subject, which Irani calls the 'entrepreneurial citizen', is constantly driven by its responsibility towards the country and its development, but only within the bounds of investor interest. This middle-class subject's role is in actively making an emerging market (hitherto made up of non-consumers) open to global capital and its financial interests.

The entrepreneurial citizen, then, is one of the more recent exploiters of *jugaad* habituation. It makes existing habituations marketable and creates a new set of habituations. It adapts indigenous demands to the interests of global financial capital (NGO and Silicon Valley investor funds), keeping national development within the expectations of global (Western) hegemonic modernity. The work performed by this citizen in becoming an expert who typifies, categorizes, and studies people to find solutions to their respective needs and concerns is reminiscent of Foucauldian governmentality (Irani 2019).

In this light, it is crucial to understand the financial regime in which KR is implicated. A project that was called volunteer driven, even though it channelled direct donations to state government coffers, is implicated in these international structures of charity, giving, and benevolence. For investors – including software and tech corporations – it offers path-breaking research into an understudied population that is developed into a portal, accounting for local specificities, language variations, topography, and bureaucratic machinations. Mobile-screen familiarity on which KR relies is used in the interest of platform capitalists. It is not my argument that this was the direct intention of the programmers or of the state government, for that matter. Rather, the ideological underpinnings of the entrepreneurial citizen already existed and implicated KR even before it was created.

Apart from this use as a potential tool of governance facilitated by the middle class, KR also has the potential to be used as a tool of surveillance. Large socio-technical systems that open vulnerable citizens to scrutiny of any kind need to be approached with caution. Surveillance is an obvious risk whenever data on citizens can be accessed from large databases, which are often interconnected. But surveillance often does not have an immediate manifestation. As her study of Aadhaar as a leaky system posits, for Henne (2019), surveillance is the key mode through which State authorities gain knowledge of their subjects. So,

although systems such as KR might make citizens visible during a disaster, the risk begins when these platforms become a part of processes that store and sell consumer data, for corporate benefit as well as for the State as and when it is demanded. Noting the frequent leaks that demonstrate Aadhaar's failings in managing citizen data, Henne (2019: 241) argues that leaky databases can make subjects the targets of structural power. In fact, she says this leakiness is becoming a 'feature of governance'.

Even if we believed that large amounts of data gathered during the floods have not been scraped and stored for future use, KR's functionalities can be deployed in the future to identify some of the most vulnerable communities. What could be the benefit of such surveillance? Coastal communities, with their distrust in the state for building infrastructure for their survival in the long term (Nagaraj 2021) and in the State's apathy when platform capitalism exploits their media habituations, might become one of the earliest groups to face acute abjection in the face of environmental distress. Despite its manifestation as need or destitution, abjection, Monahan (2017) argues, is also a recipe for becoming social and spatial misfits. This is often checked and managed through different forms of surveillance, both lateral and vertical. Monahan's analysis is primarily in the American racial and carceral context, but it explains the motivations for employing contemporary surveillance technologies.

8 Conclusion

Platform-based infrastructure is mired in the surveillance and governance roles that it plays for State and corporate actors. As discussed, KR could easily follow that legacy if some pieces changed. However, it is important to acknowledge how platforms such as KR are being appropriated within indigenous ontologies of being and living, with modes of organizing labour, resources, and bodily and affective habituations in a community in crisis, and adopting proliferating tools and networks to negotiate precarity and subalternity. These manifestations of *jugaad* go beyond the platform-centric or techno-deterministic frames of analysis. Instead, they demonstrate the techniques of modification, adaptation, and dynamic creativity that make technical systems less stagnant.

Platforms have modified and subsumed the need for and capacities of infrastructure, both materially and symbolically. In the digital sphere, they are linked to mediating everyday functions (accessing welfare, registering for vaccines, booking transportation) into a system of codes and associated practices, often impossible to circumvent. However, platforming of any kind is

predicated on networks and capacities that become habituations through repetition. They are the creative and bodily multiplicities that are often co-opted to perform on behalf of capital or another form of platforming, such as for a nation, religion, or ideology. These networks, capacities, and creative potential, which I discuss here as *jugaad*, can also assemble in ways that do not appropriate to the point of erasure. KR as a portal, as well as its associated ecologies, is an example of this platforming. Even so, the fact that it draws on an existing interpellation of entrepreneurial citizens, as well as copious data gathering, and data-based surveillance practices makes its role as a platform for profit making and governance a distinct possibility.

Meanwhile, *jugaad* ways of organizing and being constantly need to innovate in the face of unprecedented crises. In that case, platformized infrastructure offers temporary fixes, only to be undone. Although Kerala has historically been flood prone, environmental distress has been accelerated by rising sea levels, frequent cyclones, and uncharacteristic rainstorms. The problem is also compounded by local political disinterest not just in recognizing this as imminent but in actively encouraging settlement, commercial and industrial, in environmentally precarious regions (Senthilir 2018). This not only threatens the people living and working in these regions but also magnifies the problems for people who have been part of these ecologies of support and networked or affective sustenance. In other parts of the state along brackish waters, indifference to frequent floods on a smaller scale that regularly upends the lives of people, along with a drinking water problem, has caused an existential crisis that drives vast climate-based migration (Nagaraj 2021). Although residents grouse about the state's inability to build and upgrade infrastructure that can address these concerns, this problem of planetary proportions remains beyond the ability of any single state to address on its own. Can platform-based *jugaad* infrastructure address these concerns? Shome (2019) argues that no theoretical solutions exist for problems with no prior frames of reference. Although the geopolitical, colonial, and postcolonial powers might have had a role in accelerating these crises, simply addressing them will not be the solution. For Shome, artistic mediations from within communities that recognizes the depth of these issues could form part of beginning to create a way to address them.

Further, not all crises similarly mobilize networks and habituations as *jugaad* that aims to mitigate precarity. As this article was written, a global debate over the failure to keep promises about state-subsidized and patent-free intellectual rights for vaccine production is accelerating (Zaitchick 2021). The KR infrastructure, and its implicit and structural care work, offer a reference for imagining technical systems of support and organization. Moreover, it offers ways of imagining mitigation of and negotiation over large planetary

issues that call for not just the modification of tools and networks to create solutions but also resistance to commodification and co-option by 'platforms' for individual, private, or majoritarian capture.

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